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Research Article

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RESEARCH ARTICLE

16S rRNA gene-based primer pair showed high specificity and quantification accuracy in detecting freshwater Brocadiales anammox bacteria

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One sentence summary: This study proposes a slightly modified primer pair for freshwater anammox bacteria and demonstrates its applicability to a variety of molecular methods.

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ABSTRACT

Anaerobic ammonium oxidizing (anammox) bacteria are widely distributed and contribute significantly to the global nitrogen cycle. Traditionally, identification and quantification based on the 16S rRNA gene were considered not reliable because of low 16S rRNA gene sequence identity within Brocadiales. Here we hypothesize that by using appropriate primers and methodology, 16S-based detection and quantification of anammox bacteria can be accurate. We modified an existing 16S rRNA gene-based primer pair (Amx694F–Amx960R) by changing one nucleotide (Amx694F position 18, G→C) (Amx694PF–Amx960R) so that they match the sequences of most Brocadiales anammox bacteria, and evaluated the modified primer pair with 29 freshwater samples from microcosms, anammox reactors and wastewater treatment plants of various geographical origins. The primer pair showed high specificity in detection and quantification of anammox populations in samples that contained >0.1% anammox bacteria. Quantification of anammox abundance by quantitative real-time PCR and delineation of anammox species by denaturing gradient gel electrophoresis agreed well with amplicon sequencing results. A clear shift of anammox population towards '*Candidatus* Kuenenia' was observed under laboratory cultivation conditions. With the help of amplicon sequencing, we demonstrated that 16S rRNA gene-based anammox-specific primers are able to achieve qualitative and quantitative monitoring of anammox communities in wastewater treatment plants and natural freshwater environments.2007;73:5261–7.

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INTRODUCTION

Anaerobic ammonium oxidizing (anammox) bacteria play an important role in the nitrogen cycle on earth. They oxidize ammonium and reduce nitrite using the electron flow for energy conservation via a respiratory process. This synproportionation reaction of ammonium and nitrite leads to dinitrogen gas, decreasing the concentration of total nitrogen in the medium. Based on the necessity for the presence of both ammonium and nitrite in the medium, anammox bacteria appear to be specialized to live in a gradient between oxic and anoxic environmental compartments. Anammox bacteria are biotechnologically used in wastewater treatment for nitrogen removal because the anammox reaction saves operating costs on energy and electron donors compared with standard nitrogen removal procedures in which ammonium is completely oxidized to nitrate by aeration (nitrification) and successively reduced to nitrogen gas (denitrification). Intensive research has been done on anammox bacteria and anammox reactor conditions in the last two decades (Kartal *et al.* 2013). Although no single pure anammox culture has been isolated, at least 25 anammox species have been identified from mixed consortia. They are affiliated with the orders Brocadiales (genera '*Candidatus* Anammoxoglobus', '*Ca.* Brocadia', '*Ca.* Jettenia', '*Ca.* Kuenenia', '*Ca.* Scalindua', all within the family Brocadiaceae) or Planctomycetales (genus '*Ca.* Anammoximicrobium') (van Niftrik and Jetten 2012; Khramenkov *et al.* 2013).

To monitor the abundance of anammox bacteria in laboratory reactors and wastewater treatment plants, and to further explore the environmental distribution and importance of anammox bacteria, accurate detection and quantification is essential. For the detection of anammox organisms, two methods are widely used: fluorescence *in situ* hybridization (FISH) and PCR-based detection of 16S rRNA genes or functional genes. FISH protocols are work-intensive for routine application but can provide a direct view on the abundance, morphology and distribution of anammox organisms (Jetten *et al.* 2005). PCR amplification of 16S rRNA genes allows quantitative analysis in many parallel samples or in diversity studies via amplicon sequencing, but can also be combined with downstream analyses such as denaturing gradient gel electrophoresis (DGGE) or terminal restriction fragment length polymorphism analysis (tRFLP). PCR has also been used to directly target functional genes responsible for specific enzymes involved in the anammox process, e.g. hydrazine synthase (Li and Gu 2011; Harhangi *et al.* 2012).

The challenge in designing an anammox-specific primer is the relatively low 16S rRNA gene sequence identity within the anammox bacteria amounting to 87–99% sequence identity within the order Brocadiales (van Niftrik and Jetten 2012). Therefore it is difficult to achieve high coverage and high specificity with one primer pair (Zhou *et al.* 2018). The primer Pla46 (Neef *et al.* 1998), often combined with a bacterial reverse primer, amplifies *Planctomycetes* but is not specific for anammox organisms within the *Planctomycetes*. The primer pair Amx368f–Amx820r (Schmid *et al.* 2000; Schmid *et al.* 2003) has been widely used to study anammox organisms in a variety of environments but yielded poor resolution in DGGE analysis (Sonthiphand and Neufeld 2013). The primer pairs An7f–An1388r and Brod541f–Brod1260r were later described to more specifically amplify full and partial 16S rRNA genes from anammox bacteria, respectively (Penton, Devol and Tiedje 2006). The primer pair Amx694F–Amx960R amplifying only a short part of the 16S rRNA gene was used to establish a quantitative PCR (qPCR) protocol (Ni *et al.* 2010) and the primer pair A438f–A684r was proposed for DGGE and cloning (Humbert, Zopfi and Tarnawski 2012). The performance of different primers in qPCR quantification of anammox bacteria was evaluated by Sonthiphand and Neufeld (2013), but due to the lack of a corresponding amplicon sequencing analysis as a reference it is difficult to assess the reliability of qPCR quantification. In 2013 a new anammox genus '*Ca.* Anammoximicrobium moscowii' was identified and none of the above-mentioned anammox primers covers its 16S rRNA gene sequence (Khramenkov *et al.* 2013).

Here we describe our efforts to find and to evaluate specific primers for anammox bacteria applicable for amplicon sequencing, qPCR and DGGE analysis. We found bioinformatic evidence that the primer pair Amx694F–Amx960R can be optimized to match almost all currently described Brocadiales anammox species with a maximum of one mismatch. To evaluate the primer pair, we compared results of amplicon sequencing using bacterial or anammox bacteria-specific primer pairs with qPCR and DGGE results. With a wide range of different freshwater samples from complex reactor cultures to enriched anammox populations, we show that anammox abundance and population diversity can be quantitatively assessed with the new primer pair. Monitoring of anammox enrichments indicated a shift of anammox populations towards '*Ca.* Kuenenia' species in the laboratory.

MATERIAL AND METHODS

Origin of biological materials

In total, 29 samples of various types and from different locations were used in this study (Table S1, see online supplementary material). Among these 29 samples, 12 were taken from laboratory anammox reactors in Beijing (China, R1-R11) and Japan (R-Japan); 6 were taken from full-scale wastewater treatment plants in Xi'an (China, XiAnW1-XiAnW5) and Singapore (ChangiW); 7 were taken from cultures that were inoculated from reactor liquid (R1-G1, R2-G1, R4-G1, R5-G1, R-Japan-G1), wastewater treatment plant sludge (ChangiW-G1) or mangrove soil (SIN-G1); and 4 were taken from sediment-free culture lines (i.e. cultures in which no original sediment was left). Such sediment-free cultures had been transferred multiple times from a reactor (R-Japan-G6, R9-G7, R11-G3) or from the mangrove soil (SIN-G6). The number after 'G' indicates the number of consecutive transfers that had been done in batch cultures. For the 7 cultures that were only transferred once (sample name ending with G1) anammox activity was not monitored. The 4 sediment-free cultures that were transferred multiple times had anammox activity (data not shown).

Cultivation

A bicarbonate-buffered defined mineral salts medium denoted as 'anammox medium' was used for enrichment of anammox organisms. The detailed pre-medium recipe for the 'anammox medium' has been described previously as 'batch cultivation medium' (Ding, Francis and Adrian 2018) and contained 5.6 mM ammonium chloride. The pre-medium was distributed in

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100-mL aliquots to 160-mL serum bottles capped with butyl rubber stoppers and aluminum crimp caps and autoclaved. All procedures were done under anoxic conditions. After autoclaving, 0.3 mL of 1 M sterile-filtered sodium nitrite (final nitrite concentration 3 mM), and 0.2 mL sterile-filtered vitamin stock solution were injected into each bottle (Ding, Francis and Adrian 2018). Bottles were either inoculated with 10% vol/vol from a sediment-free culture with a syringe or they were opened inside an anoxic glove box (Coy laboratories) and a sludge inoculum was transferred with a sterile spatula. Cultures were incubated at 30°C in the dark without shaking.

Construction of a 16S rRNA gene sequence alignment

We downloaded 16S rRNA gene sequences from National Center for Biotechnology Information (NCBI) including 21 sequences from Brocadiales anammox strains, one sequence from '*Ca*. Anammoximicrobium moscowii' (order Planctomycetales) (Khramenkov *et al*. 2013) and 10 reference sequences (from non-anammox *Planctomycetes* or non-*Planctomycetes* strains). A ClustalW alignment was generated using the IUB DNA weight matrix, gap opening penalty of 15 and gap extension penalty of 6.66.

DNA extraction and dilution

Genomic DNA was extracted from biological samples using a column-based method as described (Chow *et al*. 2010) but omitting achromopeptidase to avoid DNA contamination from the commercial enzyme. Resulting DNA concentrations were between 0.4 and 304 ng μL^{-1} as determined with a NanoDrop 1000 spectrophotometer (Thermo Fisher). Undiluted DNA samples were sent to BGI (Shenzhen, China) for Illumina amplicon sequencing. The same DNA samples were used for PCR-DGGE and qPCR reactions after dilution as follows: DNA samples that were extracted from sludge, biofilm or sediments were diluted at least 10 times regardless of their DNA concentration to minimize PCR inhibitors; DNA samples that were extracted from sediment-free cultures were diluted if the DNA concentration was >10 ng μL^{-1}. Final DNA concentrations after dilutions were in the range 0.2–6.9 ng μL^{-1}.

Molecular analyses

qPCR was performed in a 20-μL reaction volume on an ABI 7500 FAST Real-Time PCR system. Reaction mixtures contained 10 μL of SensiFAST SYBR® Lo-ROX mixture (Bioline, UK), 0.6 μL of each forward and reverse primer (10 μM) and 2 μL of DNA template. As a quantification standard we used a plasmid containing a '*Ca*. Kuenenia stuttgartiensis' 16S rRNA gene fragment 100% matching both the bacterial primer pair 338F–518R and the anammox-specific primer pair Amx694PF–Amx960R (Table 1). Serial dilutions (10^3–10^7 copies μL^{-1}) of the plasmid DNA were used for quantification of both bacterial and anammox 16S rRNA gene copies. To ensure the accuracy of anammox abundance values, quantification of total bacteria and anammox bacteria for the same samples were always performed on the same microtiter plate with the same set of plasmid dilutions, so that errors caused by plasmid dilution or plate-to-plate variations during qPCR were eliminated. The reaction conditions were as follows: initial incubation at 95°C for 3 min, followed by 45 cycles of 95°C for 5 s and 60°C for 30 s. Melting curves were evaluated for each reaction to verify specificity of amplification. All reactions were done in technical triplicates. Nuclease-free water samples served as negative controls in all plates. Standard curves showed $r^2 > 0.9992$, efficiency of 91.1 to 93.5% for the primer pair Amx694PF–Amx960R, and $r^2 > 0.9998$, efficiency of 95.3 to 96.0% for the primer pair 338F–518R. The abundance of anammox bacteria (qPCR% in Table 2) was calculated as the ratio of anammox to bacterial 16S rRNA gene copies.

DGGE was performed with the primer pair Amx694PF–Amx960RGC (Table 1) on a DCode Universal Mutation Detection System (Bio-Rad). The PCR mixture was prepared in a 20-μL reaction volume containing 4 μL of 5 × GoTaq Flexi buffer (Promega), 2 μL of $MgCl_2$ (25 mM), 0.5 μL of dNTP mix (10 mM each), 0.67 μL of bovine serum albumin (10 mg mL^{-1}), 0.2 μL of each forward and reverse primer (10 μM), 0.1 μL of GoTaq DNA polymerase (Promega) and 1 μL of DNA template. A touch-down PCR program was used with the following conditions: initial denaturation for 5 min at 95°C; 20 cycles of 95°C for 30 s, annealing for 45 s, and 72°C for 30 s (with the annealing temperature decreasing from 65 to 55°C at 0.5°C per cycle); an additional 25 cycles with annealing at 55°C; and a 10-min final extension at 72°C. For 8 samples (XiAnW1, XiAnW2, XiAnW3, XiAnW4, XiAnW5, SIN-G1, ChangiW and ChangiW-G1), a nested approach was used using the same primer pair for a second round of amplification because the first-round PCR failed to generate enough DNA for DGGE analysis. Sample volumes for DGGE were 7–20 μL per well depending on the DNA concentration of the purified PCR samples. DGGE was run with a 10% polyacrylamide gel and a denaturing gradient of 30–75% at 120 V, 60°C for 12 h. The gel was stained for 30 min in fresh SYBR Gold nucleic acid gel stain (Invitrogen, Thermo Fisher) (5 μL SYBR Gold in 100 mL 1 × Tris-acetate-EDTA buffer) and bands were visualized under UV light using a Bio-Rad GelDoc system. A total of 65 gel bands were excised using sterile blades, and were soaked in 50 μL DNase-free water overnight to elute DNA. The eluted DNA was PCR-amplified using the primer pair Amx694PF–Amx960R. The PCR products were purified and Sanger-sequenced using the primer Amx694PF (Supplementary File S1).

Clone libraries of 16S rRNA genes were performed on samples R8, R9 and R9-G7. Previously described procedures were followed (Ding, Francis and Adrian 2018), except that the reverse primer 1541R was used for R8 and R9 and the reverse primer 1392R was used for R9-G7. The forward primer was a mixture of sequence 1 and 2 in 338F_mix in Table 1 so as to cover anammox bacteria. Sequences of 36 representative clones have been deposited in the GenBank database under accession numbers MN844039–MN844054 (R8), MN845211–MN845223 (R9), MN844055–MN844061 (R9-G7).

BlastN (version 2.6.0) was used to compare sequences obtained from Sanger sequencing of DGGE bands or selected operational taxonomic unit (OTU) sequences from the amplicon sequencing datasets (see below) against the NCBI nucleotide (nt) database.

Amplicon sequencing

Undiluted DNA samples of all 29 biological samples were sent on dry ice to BGI (Shenzhen, China), and all subsequent steps (DNA dilution, PCR, library construction, sequencing) were performed by BGI. The DNA samples were subjected to barcoded primer PCR amplification using the anammox-specific primer pair Amx694AF–Amx960R and the bacterial primer pair 338F–806R (V3–V4 region) (Table 1). PCR and libraries were performed for all samples individually using barcoded primers and were pooled in equimolar concentrations and the two pools were used

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Table 1. Primers mentioned in this study.

Primer	Sequence (5′-3′)	Comment	Reference
Amx368f	TTC GCA ATG CCC GAA AGG	For *in silico* comparison	Schmid *et al.* (2003)
Amx820r	AAA ACC CCT CTA CTT AGT GCC C	For *in silico* comparison	Schmid *et al.* (2000)
A438f	GTC RGG AGT TAD GAA ATG	For *in silico* comparison	Humbert, Zopfi and Tarnawski (2012)
A684r	ACC AGA AGT TCC ACT CTC	For *in silico* comparison	Humbert, Zopfi and Tarnawski (2012)
Amx694F	GGG GAG AGT GGA ACT TCG G	For *in silico* comparison	Ni *et al.* (2010)
Amx694PF	GGG GAG AGT GGA ACT TCT G	Anammox-specific forward primer for qPCR and DGGE	This study
Amx694AF	GGG GAG ART GGA ACT TCT G	Anammox-specific forward primer for amplicon sequencing	This study
Amx694PFGC	(GC-clamp)-GGG GAG AGT GGA ACT TCT G	Amx694PF plus 40-bp GC clamp at 5' (CGC CCG CCG CGC GCG GCG GGC GGG GCG GGG GCA CGG GGG G), for DGGE	This study
Amx960R	GCT CCA CCG CTT GTG CGA GC	Anammox-specific reverse primer for qPCR and amplicon sequencing	Ni *et al.* (2010)
Amx960RGC	(GC-clamp)-GCT CCA CCG CTT GTG CGA GC	Amx960R plus 40-bp GC clamp at 5' (CGC CCG CCG CGC GCG GCG GGC GGG GCG GGG GCA CGG GGG G), for DGGE	This study
338F	ACT CCT ACG GGA GGC AGC AG	Bacterial forward primer for qPCR and amplicon sequencing	Lane (1991)
338F_mix	Mixture of (1) ACT CCT ACG GGA GGC AGC AG (2) ACT CCT ACG GGA GGC TGC AG (3) ACT TCT ACG GAA GGC TGC AG (4) ACA CCT ACG GGT GGC TGC AG (5) ACA CCT ACG GGT GGC AGC AG	Proposed bacterial forward primer for qPCR and amplicon sequencing (1) original 338F, (2) covers most Brocadiales anammox bacteria except '*Ca.* Scalindua' but includes '*Ca.* Scalindua wagneri', (3) covers '*Ca.* Scalindua' anammox bacteria except '*Ca.* Scalindua wagneri', (4) and (5) are modified based on probes EUB338II and EUB338III (Daims *et al.* 1999)	This study
518R	ATT ACC GCG GCT GCT GG	Bacterial reverse primer for qPCR	Muyzer, Dewaal and Uitterlinden (1993)
518R_mix	Mixture of (1) ATT ACC GCG GCT GCT GG (2) TTT ACC GCG GCT GCT GG (3) CTT ACC GCG GCT GCT GG (4) GTT ACC GCG GCT GCT GG (5) ATG ACC GCG GCT GCT GG (6) ATC ACC GCG GCT GCT GG	Proposed bacterial reverse primer for qPCR (1) original 518R, (2–6) variants of 518R as shown in Table S9	This study
806R	GGA CTA CHV GGG TWT CTA AT	Bacterial reverse primer for amplicon sequencing	Caporaso *et al.* (2012)
1392R	ACG GGC GGT GTG TRC	Bacterial reverse primer for clone library of R9-G7	Lane (1991)
1541R	AAG GAG GTG ATC CAG CCG CA	Bacterial reverse primer for clone libraries of R8 and R9	Löffler *et al.* (2000)

for two separate Illumina MiSeq sequencing runs (300 bp read length, paired-end reads). The two resulting datasets will be referred to as 'AMX' (for the dataset from the anammox-specific primer pair) and 'BAC' (for the dataset from the bacterial primer pair).

Paired-end MiSeq reads were assembled using the fastq_mergepairs option from the USEARCH software package version 10.0.240 (Edgar 2010) with a minimum identity of 80% and a maximum mismatch number of 40 bp for overlapping regions. Primers were trimmed from both edges of the assembled read pairs using fastx_truncate from USEARCH. Quality filtering was applied using fastq_filter from USEARCH with a maximum expected error number of 1.0. After filtering, there were between 48 850 and 287 957 clean reads for each sample in the 'AMX' dataset, and between 6021 and 151 157 clean reads for each sample in the 'BAC' dataset. Unique sequences were picked using fastx_uniques from USEARCH. OTU picking from unique sequences was done at a clustering identity of 100% (meaning that OTUs are only clustered if they are 100% identical and even a single mismatch leads to a new OTU) using the unoise3 option from USEARCH (Edgar 2016) and chimeras were removed during this step. OTU tables were generated using the otutab option from USEARCH.

Taxonomies were assigned to OTUs using three taxonomy databases—SILVA (Yilmaz *et al.* 2014), Greengenes (McDonald *et al.* 2012) and RDP (Wang *et al.* 2007). SILVA and Greengenes assignments were performed using the script assign_taxonomy (minimum identity of 90% for assignment) from Quantitative

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Table 2. Overall diversity and relative shares of anammox bacteria in the 29 investigated samples.

Category	Sample label	Sample type	Shannon index[a]		Anammox bacteria abundance		518R %[d]
			Original	Norm	qPCR%[b]	Amplicon%[c]	
Samples from active laboratory anammox reactors	R1	Sludge	4.31	4.26	10.9	5.0	27.0
	R2	Granule	3.59	3.56	7.1	0.83	80.6
	R3	Biofilm on the fillings	4.55	4.53	3.6	0.29	36.5
	R4	Granule	4.32	4.29	8.3	3.6	49.6
	R5	Granule	4.11	4.05	28.1	23.3	39.4
	R6	Biofilm on the fillings	2.71	2.67	24.0	0.52	86.6
	R7	Biofilm on the fillings	1.37	1.33	16.6	0.16	97.3
	R8	Granule	2.96	2.93	11.4	2.5	49.9
	R9	Granule	1.05	1.04	6.1	0.25	97.5
	R10	Granule	3.19	3.16	6.1	5.8	36.3
	R11	Granule	3.35	3.32	8.1	0.49	67.6
	R-Japan	Granule	3.91	3.85	1.7	0.23	57.7
Samples from full-scale wastewater treatment plant	ChangiW	Sludge	4.87	4.79	0.13	0	17.8
	XiAnW1	Sludge	5.62	5.51	0.22	0.04	18.9
	XiAnW2	Biofilm on the fillings	4.88	4.81	0.02	0.02	44.6
	XiAnW3	Biofilm on the fillings	5.23	5.16	0.05	0	27.1
	XiAnW4	Sludge	6.05	6.05	0.25	0	18.8
	XiAnW5	Biofilm on the fillings	5.00	4.96	0.03	0	35.5
Culture after one transfer in anammox medium	SIN-G1	Microcosm	4.63	4.62	0.75	0.25	8.8
	R-Japan-G1	Sediment-free culture	3.81	3.75	0.44	0.13	24.3
	ChangiW-G1	Culture with sediment	2.84	2.79	0.13	0.04	13.1
	R1-G1	Culture	4.82	4.80	9.4	4.2	18.1
	R2-G1	Culture	3.44	3.40	18.3	16.0	56.2
	R4-G1	Culture	3.39	3.35	1.4	0.37	1.6
	R5-G1	Culture	4.15	4.09	16.0	5.8	52.4
Culture after multiple transfers in anammox medium	SIN-G6	Sediment-free culture	3.53	3.51	18.3	3.8	22.4
	R-Japan-G6	Sediment-free culture	2.28	2.27	12.9	24.2	0.3
	R9-G7	Sediment-free culture	1.06	1.06	19.5	24.0	0.1
	R11-G3	Sediment-free culture	2.21	2.19	8.3	17.1	0.4

[a]Calculated based on the 'BAC' dataset. Original: calculated with the original sample sizes. Norm: calculated with normalized sample size of 6021 equalling to the smallest sample size for XiAnW4 (bootstrap: 100).
[b]The abundance of anammox bacteria as quantified by qPCR in technical triplicates using the anammox-specific primer pair Amx694PF–Amx960R and the bacterial primer pair 338F–518R. Standard deviations were maximum ~10% of the values shown and are not displayed.
[c]The abundance of anammox bacteria as quantified by amplicon sequencing using the bacterial primer pair 338F–806R ('BAC' dataset).
[d]Abundance of OTUs that have mismatches with the qPCR reverse primer 518R, as shown in Table S9.

Insights into Microbial Ecology (QIIME) version 1.9.1 (Caporaso *et al.* 2010). For SILVA, SILVA128 representative 16S rRNA gene sequence data set (clustering at 99% identities) was used as reference sequences and the 'majority_taxonomy_7_levels' taxonomy string file was used as the mapping file (both files were from the QIIME-compatible SILVA 128 ribosomal RNA gene database). For Greengenes, taxonomy associated with the database version 13_8 (clustering at 99% identities) was used. RDP classification was done on the online server (rdp.cme.msu.edu) using RDP Naive Bayesian rRNA Classifier Version 2.11 and RDP 16S rRNA training set 16 (bootstrap threshold 80% for 'BAC' and 50% for 'AMX'). Consensus taxonomy was determined if at least two of the three taxonomy classifications agreed with each other.

For the 'BAC' dataset, the abundance of anammox bacteria (Amplicon% in Table 2) was calculated as the ratio of anammox-affiliated OTU count to total OTU count. Shannon index for the original OTU table (column 'Original' in Table 2) was calculated for each sample using the alpha_div option from USEARCH (natural logarithm base). Shannon index for normalized sample sizes (using the smallest sample size 6021) was calculated using the 'vegan' package in R with a bootstrap value of 100 (column 'Norm' in Table 2).

Anammox-affiliated OTUs were defined as OTUs belonging to any of the known anammox genera according to taxonomy assignment in the aforementioned step. For both datasets, the Bray–Curtis dissimilarity matrices showing OTU composition differences between samples were calculated using the beta_div option from USEARCH.

Raw Illumina sequencing reads were deposited to NCBI Sequence Reads Archive (BioProject accession: PRJNA587599, BioSample accession: SAMN13195819–SAMN13195847, Run accession: SRR10418783–SRR10418840).

RESULTS

In silico evaluation of anammox-specific 16S rRNA gene primer pairs

We first generated an alignment of 16S rRNA gene sequences from currently defined anammox species and aligned three previously described primer pairs to this alignment (Table S2, see online supplementary material): Amx368f–Amx820r (Schmid *et al.* 2000; Schmid *et al.* 2003), A438f–A684r (Humbert, Zopfi and Tarnawski 2012) and Amx694F–Amx960R (Ni *et al.* 2010) (Table 1).

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We noted that the forward primer Amx694F (5′-GGG GAG AGT GGA ACT TCG G-3′) had a 1-bp mismatch near the 3′ terminus with all of the available anammox sequences (highlighted guanine, position 18), and therefore revised it as Amx694PF (5′-GGG GAG AGT GGA ACT TCT G-3′) (letter 'P' for 'partial'). The revised primer pair Amx694PF–Amx960R had the lowest number of total mismatches among all primer pairs and perfectly matched all Brocadiales anammox genera except *Scalindua* with one mismatch. Substitution of a guanine residue at position 8 to a degenerate base (R, indicating A or G) led to the primer Amx694AF (5′-GGG GAG ART GGA ACT TCT G-3′) (letter 'A' for 'all'). This primer matched perfectly with all Brocadiales anammox 16S rRNA gene sequences except the 16S rRNA gene from the single species *Scalindua marina*. While Amx960R matches the 16S rRNA gene sequences of many *Planctomycetes*, *Actinobacteria* and *E. coli* with one or two mismatches, Amx694PF and Amx694AF have a minimum of three and an average of four to five mismatches with all aligned non-anammox 16S rRNA gene sequences. The newly proposed primer pair Amx694AF/Amx694PF–Amx960R covers a region of about 280 bp in the 16S rRNA gene sequence, and is therefore potentially suitable for qPCR and DGGE.

16S rRNA gene amplicon sequencing using the bacterial primer pair 338F–806R

Total DNA was extracted from all 29 samples and the 16S rRNA genes were amplified with the universal bacterial primer pair 338F–806R. Amplicons were sequenced on an Illumina MiSeq platform and assembled to the 'BAC' dataset to provide an overview of the community structure and values of anammox abundance as references for qPCR and DGGE experiments. In addition, the phylogenetic information of the anammox population was used for comparison with results from amplifications with anammox-specific primers. The 'BAC' dataset was composed of a total of 3828 unique OTUs from all 29 samples together (sequences provided in fasta-format in Supplementary File S2; abundance and taxonomic affiliations in Table S3, see online supplementary material). The results were grouped according to sample origin from reactors, wastewater treatment plants, single transfers or multiple transfers (Table 2). Among the 29 samples the highest α-diversity was found in samples from wastewater treatment plants (average Shannon index H = 5.3), followed by cultures after single transfer in anammox medium (H = 3.9), cultures from laboratory anammox reactors (H = 3.3) and cultures after multiple transfers in anammox medium (H = 2.3) (Table 2). Microbial community structures among the four sample groups were vastly different, whereas remarkable similarities were observed between samples within the same group (Fig. 1A). Dominant species were *Chloroflexi* (in most laboratory anammox reactor samples), *Bacteroidetes* (in wastewater treatment plant samples) and *Proteobacteria* (in all sample categories).

By calculating Bray–Curtis dissimilarity between samples and clustering the results, we found that cultures that were transferred once from active anammox reactors into anammox medium (R1-G1, R2-G1, R5-G1, R-Japan-G1, ChangiW-G1) showed moderate similarity to their mother cultures except R4-G1 (Table S4, Figure S1A, see online supplementary material). Cultures after multiple transfers in anammox medium clustered together, and samples from wastewater treatment plants also clustered together (Figure S1A). Among samples from wastewater treatment plants, samples from the same location taken at different dates (XiAnW1/XiAnW4, XiAnW2/XiAnW5, see Table S1 for sampling dates) also clustered together (Table S4, Figure S1A).

Anammox bacteria abundances between 0 and 24% were found across samples (Table 2) with the lowest observed for wastewater treatment plants (0–0.04%) and highest for cultures after multiple transfers (3.8–24.2%). Low abundances of anammox bacteria in the wastewater treatment plants were expected because the prominent nitrogen removal process in those wastewater plants was denitrification. In anammox reactors, anammox bacteria abundance of 0.16 to 23.3% was found. Among the singly transferred cultures, R2-G1 showed an increase in anammox bacteria abundance from 0.83% in R2 to 16.0% in R2-G1.

In our samples, *Pseudomonas* species were abundant in multiple-transferred R-Japan-G6 (42.9%, of which 40.3% were *Pseudomonas stutzeri* OTU number BAC0008), R9-G7 (69.7%, of which 68.2% were *P. stutzeri* BAC0008) and R11-G3 (74.3%, of which 48.8% were *Pseudomonas bauzanensis* BAC0010/BAC0057 and 19.7% were *Pseudomonas xiamenensis* BAC0033/BAC0066), as well as in singly transferred R4-G1 (42.7%, of which 38.0% were *P. stutzeri* BAC0002/BAC0218), while in all other samples *Pseudomonas* abundance was <5% (Table S5, see online supplementary material). *P. stutzeri* (BAC0008) also increased in R1-G1 (1.4%, compared with 0.04% in R1) and in SIN-G6 (4.0%, compared with 0.4% in SIN-G1).

16S rRNA gene amplicon sequencing using the anammox-specific primer pair Amx694AF–Amx960R

The DNA samples, from which the 'BAC' dataset was generated, were also used for amplicon sequencing with the modified anammox-specific primer pair Amx694AF–Amx960R to generate the 'AMX' dataset. A total of 306 unique OTUs were identified in the 'AMX' dataset from all 29 samples together (sequences provided in fasta format in Supplementary File S3; abundance and taxonomic affiliations in Table S5). '*Ca*. Kuenenia', '*Ca*. Brocadia', '*Ca*. Anammoxoglobus' and '*Ca*. Jettenia'-affiliated OTUs were detected, but no '*Ca*. Scalindua' anammox bacteria were detected (Figure S2, Table S6, see online supplementary material). In 22 of the 29 samples in the 'AMX' dataset, only low abundances of non-anammox affiliated OTUs were detected up to an abundance of 1.2%. In the remaining 7 samples, which included all samples from wastewater treatment plants XiAnW1, XiAnW2, XiAnW3, XiAnW4, XiAnW5, ChangiW and the first-generation transfer ChangiW-G1, Actinobacteria and Planctomycetes OM190 were abundantly detected (Figs. 1B and 2, and Table S5). As revealed by the 'BAC' dataset, these 7 samples contained the lowest abundance of anammox bacteria among the 29 samples: no anammox sequences were detected in 4 out of the 7 samples and <0.04% in the other 3 samples. In contrast, all other 22 samples showed anammox bacteria abundances of ≥0.13% (calculated based on Table S3). Therefore, ~0.1% of anammox bacteria (0.04 to ~0.13% of the total population) appear to be required for specificity of this primer pair.

The clustering of tree topologies comparing the population similarities between samples were very similar for the two datasets 'AMX' and 'BAC' (Figure S1). Samples after single transfers in anammox medium mostly clustered together with their mother cultures and distinct anammox populations were kept (Figure S1B, Table S7, see online supplementary material). Due to the ~100 bp overlap of amplified regions of the 'AMX' and 'BAC' datasets, we were able to associate anammox-affiliated OTUs in

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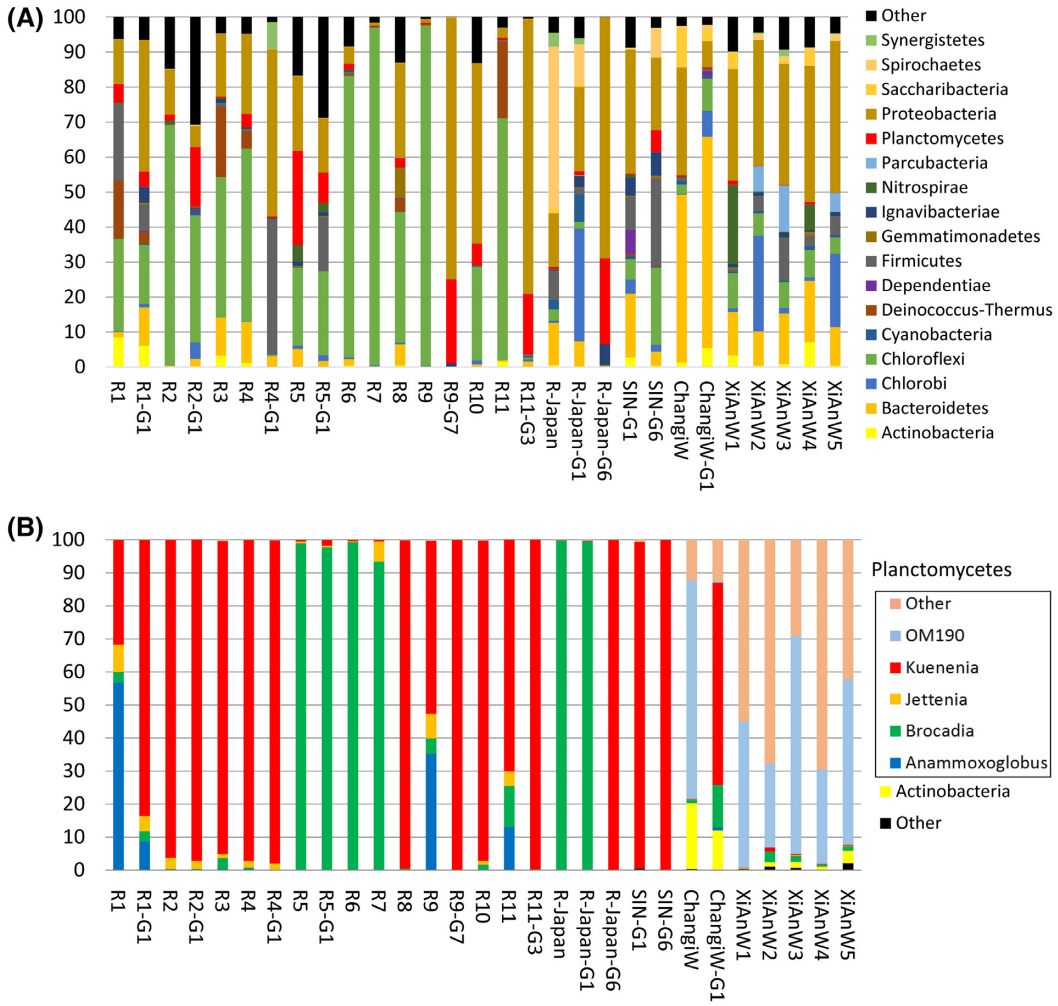


Figure 1. Phylogenetic composition of samples in this study. (**A**) Distribution of phyla after amplicon sequencing with the bacterial primer pair 338F–806R. (**B**) Amplicon sequencing with anammox-specific primer pair Amx694AF–Amx960R. If an OTU is affiliated with several anammox genera, the OTU counts are shared among these genera.

these two datasets (Table S7) and found good agreement in relative abundances of OTUs (Table S7).

All samples showed strong enrichment towards '*Ca*. Kuenenia' species after multiple transfers in anammox medium although '*Ca*. Anammoxoglobus' and '*Ca*. Brocadia' were among the major anammox populations in the parent cultures R9, R11 and R-Japan. Whereas samples originating from the anammox reactors (R9-G7, R11-G3, R-Japan-G6) contained the same '*Ca*. Kuenenia' OTU (AMX001 or BAC0003, 100% sequence identity with '*Ca*. Kuenenia stuttgartiensis' strain MBR1), SIN-G6 which was enriched from mangrove soil in Singapore contained a different but highly similar '*Ca*. Kuenenia' OTU (AMX006 or BAC0227/BAC0398, 99% identity with '*Ca*. Kuenenia stuttgartiensis' strain MBR1). Separation of such highly similar OTUs was made possible by setting the clustering threshold to 100% during OUT picking (see Methods section) (Edgar 2016; Edgar 2017).

Quantification of anammox bacteria abundance with qPCR

To establish a simple and accurate method to monitor anammox bacteria abundance in complex samples, qPCR reactions using the bacterial primer pair 338F–518R and the anammox-specific primer pair Amx694PF–Amx960R were performed on the same DNA samples for which the above described amplicon sequences were determined. Population analyses in the 'AMX' dataset already showed that the primer pair Amx694AF–Amx960R was specific for anammox populations in 22 samples (except ChangiW-G1, ChangiW, XiAnW1–XiAnW5). Indeed, anammox bacteria abundance values in these 22 samples determined by qPCR agreed well with the abundance values obtained by 'BAC' amplicon sequencing (Table 2). Anammox bacteria abundance values quantified by qPCR were generally higher than those obtained from amplicon sequencing, except in three out of four samples from highly enriched laboratory cultures (R-Japan-G6, R9-G7 and R11-G3). Since the specificity of the primer pair Amx694PF–Amx960R towards anammox bacteria in these 22 samples had been verified by amplicon sequencing, we hypothesized that the discrepancy was due to mismatches in the bacterial primer pair 338F–518R with major populations in the samples. On one hand, the forward primer 338F shows a mismatch with all Brocadiales anammox genera except '*Ca*. Scalindua' (Table S8, see online supplementary material). On the other hand, the reverse primer 518R shows mismatches with 1032

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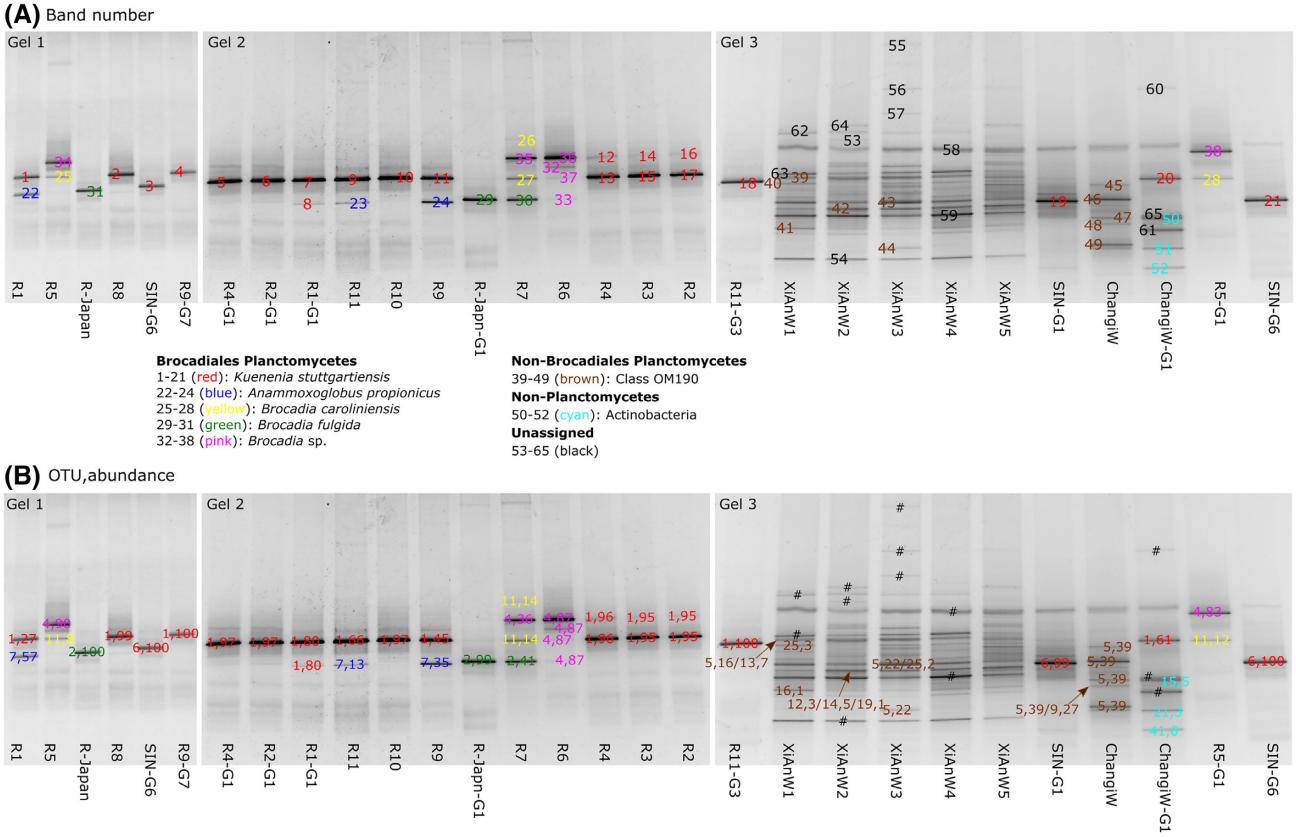


Figure 2. Denaturing gradient gel electrophoresis (DGGE) profiles of 28 samples using the primer pair Amx694PF–Amx960RGC. (**A**) Gel bands with band numbers and affiliation based on NCBI nt BLAST result. (**B**) Gel bands with associated OTU numbers in the 'AMX' dataset and their respective abundance in percentage (rounded to integer): e.g. '1,97' indicates the band can be associated with the OTU AMX001 and abundance of 97%. Association criteria are >180 bp sequence coverage and >95% identity. Bands that cannot be associated to OTUs are labeled with '#'. Labels are directly on top of the bands unless indicated with an arrow.

out of 3828 OTUs in the 'BAC' dataset which account for 0.1%-97% of the detected population (Table S9, see online supplementary material). The mismatch in primer 338F probably led to an underestimation of anammox abundance in the 'BAC' dataset, and mismatches in both 338F and 518R probably led to an underestimation of the total bacteria number and therefore an overestimation of the anammox abundance values in qPCR quantification. Particularly, in samples R2, R6, R7 and R9, between 81 and 97% of the retrieved amplicons contain mismatches to the reverse primer 518R (Table 2, Table S9). The high abundances of mismatched sequences probably caused the discrepancies in anammox abundance values by qPCR and amplicon sequencing in these samples.

16S rRNA clone libraries on three samples (R8, R9 and R9-G7) also supported the hypothesis that primer binding is the cause of discrepancies in anammox abundance values (Table S10, see online supplementary material). The forward primer set for clone libraries was a mixture of sequences 1 and 2 from the 338F_mix (Table 1) which then covered the anammox genera. The reverse primers were 1541R for R8 and R9, and 1392R for R9-G7. It turned out that abundance values of anammox-affiliated clones in R8 and R9 were higher than those determined by amplicon sequencing and even those by qPCR. Since anammox bacterial 16S rRNA sequences match perfectly to both the forward and reverse primers, the overestimation is probably caused by poor alignment of 1541R to non-anammox major populations in R8 and R9.

Two new primer mixtures 338F_mix and 518R_mix were then proposed for better coverage of bacteria including all Brocadiales anammox species (Table 1). These two primer mixtures can be used in future instead of 338F and 518R to target bacteria in samples by qPCR and amplicon sequencing.

Anammox species delineation with DGGE

A DGGE protocol with the anammox-specific primer pair Amx694PF–Amx960R was developed (see Methods section). A 40-bp GC clamp was attached to the 5′-terminus of Amx960R (denoted Amx960RGC) (Table 1) because better DGGE band separation was obtained with such a GC clamp at the primer Amx960R than with a GC clamp at the primer Amx694FP (Figure S3, see online supplementary material). The DGGE protocol was applied to the same DNA samples that were used for amplicon sequencing and qPCR (except R-Japan-G6). Results demonstrated the fine resolving ability of DGGE (Fig. 2, Table S11, see online supplementary material). Even similar anammox OTUs were well separated on the DGGE gel. For example, the two highly similar '*Ca*. Kuenenia stuttgartiensis' OTUs AMX001 (dominant band in R1, R8 and R9-G7) and AMX006 (dominant band in SIN-G6) with only 2 bp differences, were well separated (Gel 1 in Fig. 2). In samples R6 and R7, four different anammox OTUs were separated (Gel 2 in Fig. 2).

Species delineation and relative abundance obtained by DGGE agreed well with those determined by amplicon sequenc-

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ing in the 'AMX' dataset (Table S6). The major bands in all samples matched perfectly with the 'AMX' dataset in terms of species delineation and relative abundance (Fig. 2).

Eight samples (XiAnW1–XiAnW5, SIN-G1, ChangiW and ChangiW-G1) showed no visible bands on the PCR agarose gel after one round of PCR, and therefore a second round of PCR was done with the same primer pair using the first round PCR products as templates in order to generate enough DNA for DGGE. All the DGGE gel bands that were not affiliated with anammox species belong to these eight samples that went through two rounds of PCR amplification (Table S11). Moreover, all these eight samples have a ghost band at the same position that could not be assigned (band No. 58, Table S11). Except for SIN-G1 and ChangiW-G1 where bands affiliated with '*Ca*. Kuenenia' anammox were identified, no other anammox-affiliated bands were identified in these eight samples.

DISCUSSION

In this study, a 16S rRNA gene-targeted anammox-specific primer pair Amx694PF–Amx960R (or Amx694AF–Amx960R if a wobble position is included) was designed and extensively evaluated in complex biological samples through amplicon sequencing, qPCR and DGGE, showing accurate detection and quantification of Brocadiales anammox bacteria.

Anammox-specific 16S rRNA gene-based markers were sometimes considered as being insufficiently specific, compared with functional gene markers such as those targeting hydrazine synthase genes (*hzs*), hydrazine dehydrogenase genes (*hzo*) or nitrite oxidoreductase genes (*nir*) in anammox bacteria (Harhangi *et al*. 2012; Wang *et al*. 2016). Here we showed that with the newly designed primers some of the obstructions can be overcome and reliable quantitative monitoring can be achieved. Through amplicon sequencing, we demonstrated that 16S rRNA gene-based detection with the proposed primer pair was specific for anammox bacteria in samples with >0.1% anammox bacteria. Abundance of anammox bacteria >0.1% occurs in a wide range of samples (Moore *et al*. 2011; Sonthiphand and Neufeld 2013). For samples with lower anammox bacteria abundance, DGGE can be used to analyse the specificity of the primer pairs, as was also proposed in this study. Compared with functional gene markers, 16S rRNA gene markers possess multiple advantages. Functional genes among anammox organisms from different genera are more variable in sequence than 16S rRNA genes. To account for such variations, functional gene primers often contain multiple degenerate bases (Wang *et al*. 2016; Park *et al*. 2017). Degenerate bases in the primers, however, can cause unspecific amplification and yield inconsistent results (Tsushima, Kindaichi and Okabe 2007). The primer pair that we here proposed contains no (Amx694PF) or only one (Amx694AF) degenerate base. Another advantage of anammox-specific 16S rRNA gene-targeted primers is that they allow quantification of anammox bacteria abundance in the total population when used together with bacterial primers, as shown in our study.

Many previous studies have employed qPCR techniques to quantify anammox bacteria using 16S rRNA gene-based anammox-specific primers (Li *et al*. 2010; Ni *et al*. 2010; Moore *et al*. 2011; Humbert, Zopfi and Tarnawski 2012; Qiao *et al*. 2013; Sonthiphand and Neufeld 2013; Wenk *et al*. 2013; Zheng *et al*. 2016; Narita *et al*. 2017; Park *et al*. 2017; Zhang *et al*. 2017). Among them, two studies used anammox-specific (Amx368f–Amx820r and A438f–A684r, respectively) and bacterial primers (338F–518R) for quantification (Moore *et al*. 2011; Sonthiphand and Neufeld 2013), and therefore anammox bacteria abundances in the total population could be calculated. Here we further developed this approach by proposing the anammox primer as described above and by using the same plasmid standard for both anammox bacteria and total bacteria. The abundance quantification by qPCR in our study agreed well with amplicon sequencing results, and the discrepancies can be explained by mismatches with the primers used for bacterial quantification (338F and 518R).

Several studies have employed DGGE to fingerprint anammox communities in the past. Some of them used a nested PCR approach, using anammox-specific primer for a first amplification and bacterial primer for a second amplification, before amplicons were analysed by DGGE (Moore *et al*. 2011; Sonthiphand and Neufeld 2013). However, the nested approach with the primer pair An7F–An1388R was found unsuitable for environmental samples with low anammox bacteria abundances (Sonthiphand and Neufeld 2013). Anammox-specific DGGE with only one round of PCR with anammox-specific primers, e.g. Pla46F–Amx368R or Amx368f–Amx820r (GC clamp position was not indicated) (Hsu *et al*. 2014), as well as A438fGC–A684r or Amx368fGC–Amx820r (Sonthiphand and Neufeld 2013) showed successful separation of anammox species. In our study, the primer pair Amx694PF–Amx960RGC showed both high resolving ability and high specificity to anammox bacteria in all samples that yielded sufficient PCR products after one round of PCR (Fig. 2). Given the high *in silico* coverage of our primer pair Amx694PF–Amx960RGC, we expect that this primer pair is suitable for applications that require a comprehensive coverage of freshwater anammox bacteria.

Through amplicon sequencing and DGGE, we identified four genera of anammox bacteria—'*Ca*. Kuenenia', '*Ca*. Brocadia', '*Ca*. Jettenia' and '*Ca*. Anammoxoglobus' in our samples. No '*Ca*. Scalindua' anammox bacteria were found, which is in line with previous findings that '*Ca*. Scalindua' only survive in medium with salinity >1.5% salt (wt/vol) (Oshiki *et al*. 2015). We also found that in most of the transferred cultures the relative abundance of '*Ca*. Kuenenia stuttgartiensis' increased, whereas the relative abundance of the originally abundant '*Ca*. Anammoxoglobus' and '*Ca*. Brocadia' decreased after successive transfers, as seen in R1-G1 (from R1), R9-G7 (from R9), R11-G3 (from R11) and R-Japan-G6 (from R-Japan) (Table S5). This indicates that our cultivation conditions specifically enriched for '*Ca*. Kuenenia' anammox bacteria.

In some of the subcultures (R4-G1, R-Japan-G6, R9-G7 and R11-G3) we observed specific enrichment of *Pseudomonas* species, mainly affiliated with *P. stutzeri* and *P. xiamenensis* which are known denitrifying bacteria (Lalucat *et al*. 2006; Lai and Shao 2008). Enrichment of denitrifying bacteria during sub-transfers was not surprising given the presence of 3 mM nitrite in the medium, and their occurrence in anammox cultures has been reported before (Jetten *et al*. 1999). However, it is not clear why *Pseudomonas* denitrifiers are specifically enriched in the subcultures, considering the large diversity of denitrifiers. The enrichment of *Pseudomonas* seemed to have a positive impact on anammox abundance quantification: in samples where *Pseudomonas* was abundant, OTUs that perfectly matched the reverse qPCR primer 518R amounted to >98% (Table S9), and therefore qPCR quantification of total bacteria and anammox abundance was considered more accurate in these samples.

In summary, two new anammox-specific primer pairs were devised and validated in this study. The primer pair Amx694PF–

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Amx960R matches perfectly all Brocadiales anammox bacteria except the genus '*Ca*. Scalindua'. The primer pair Amx694AF–Amx960R containing one wobble position matches perfectly all Brocadiales anammox bacteria except '*Ca*. Scalindua marina'. The primer pairs were highly specific to anammox bacteria, as demonstrated by amplicon sequencing, qPCR and DGGE on a variety of biological samples. Results obtained with the primer pairs helped to demonstrate successful cultivation of sediment-free anammox cultures in serum bottles with the described anammox medium. Through comparison of samples, we found that our anammox cultivation conditions enriched '*Ca*. Kuenenia' anammox bacteria as well as putative denitrifying bacteria (*Pseudomonas*).

SUPPLEMENTARY DATA

Supplementary data are available at *FEMSEC* online.

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Conflict of interest. None declared.

REFERENCES

Caporaso JG, Lauber CL, Walters WA *et al*. Ultra-high-throughput microbial community analysis on the Illumina HiSeq and MiSeq platforms. *ISME J* 2012;**6**:1621–4.

Caporaso JG, Kuczynski J, Stombaugh J *et al*. QIIME allows analysis of high-throughput community sequencing data. *Nat Methods* 2010;**7**:335–6.

Chow WL, Cheng D, Wang S *et al*. Identification and transcriptional analysis of *trans*-DCE-producing reductive dehalogenases in *Dehalococcoides* species. *ISME J* 2010;**4**:1020–30.

Daims H, Brühl A, Amann R *et al*. The domain-specific probe Eub338 is insufficient for the detection of all bacteria: Development and evaluation of a more comprehensive probe set. *Syst Appl Microbiol* 1999;**22**:434–44.

Ding C, Francis EO, Adrian L. Anaerobic ammonium oxidation (anammox) with planktonic cells in a redox-stable semi-continuous stirred-tank reactor. *Environ Sci Technol* 2018;**52**: 5671–81.

Edgar RC. Search and clustering orders of magnitude faster than BLAST. *Bioinformatics* 2010;**26**:2460–1.

Edgar RC. UNOISE2: Improved error-correction for Illumina 16S and ITS amplicon sequencing. *bioRxiv* 2016;081257.

Edgar RC. Updating the 97% identity threshold for 16S ribosomal RNA OTUs. *bioRxiv* 2017;192211.

Harhangi HR, Le Roy M, van Alen T *et al*. Hydrazine synthase, a unique phylomarker with which to study the presence and biodiversity of anammox bacteria. *Appl Environ Microbiol* 2012;**78**:752–8.

Hsu S-C, Lai Y-C, Hsieh P-H *et al*. Successful enrichment of a rarely found *Candidatus Anammoxoglobus propionicus* from leachate sludge. *J Microbiol Biotechnol* 2014;**24**:879–87.

Humbert S, Zopfi J, Tarnawski S-E. Abundance of anammox bacteria in different wetland soils. *Environ Microbiol Rep* 2012;**4**:484–90.

Jetten M, Schmid M, van de Pas-Schoonen K *et al*. Anammox organisms: Enrichment, cultivation, and environmental analysis. In: Jared RL (ed). *Methods Enzymol*. New York: Academic Press, 2005, 34–57.

Jetten MS, Strous M, Van de Pas-Schoonen KT *et al*. The anaerobic oxidation of ammonium. *FEMS Microbiol Rev* 1999;**22**: 421–37.

Kartal B, de Almeida NM, Maalcke WJ *et al*. How to make a living from anaerobic ammonium oxidation. *FEMS Microbiol Rev* 2013;**37**:428–61.

Khramenkov SV, Kozlov MN, Kevbrina MV *et al*. A novel bacterium carrying out anaerobic ammonium oxidation in a reactor for biological treatment of the filtrate of wastewater fermented sludge. *Microbiology* 2013;**82**:628–36.

Lai Q, Shao Z. *Pseudomonas xiamenensis* sp. nov., a denitrifying bacterium isolated from activated sludge. *Int J Syst Evol Microbiol* 2008;**58**:1911–5.

Lalucat J, Bennasar A, Bosch R *et al*. Biology of *Pseudomonas stutzeri*. *Microbiol Mol Biol Rev* 2006;**70**:510–47.

Lane DJ. 16S/23S rRNA sequencing. In: Stackebrandt E, Goodfellow M (eds). *Nucleic acid techniques in bacterial systematics*. New York: Wiley, 1991, 115–75.

Li H, Chen S, Mu B-Z *et al*. Molecular detection of anaerobic ammonium-oxidizing (anammox) bacteria in high-temperature petroleum reservoirs. *Microb Ecol* 2010;**60**: 771–83.

Li M, Gu J-D. Advances in methods for detection of anaerobic ammonium oxidizing (anammox) bacteria. *Appl Microbiol Biotechnol* 2011;**90**:1241–52.

Löffler FE, Sun Q, Li JR *et al*. 16S rRNA gene-based detection of tetrachloroethene-dechlorinating *Desulfuromonas* and *Dehalococcoides* species. *Appl Environ Microbiol* 2000;**66**: 1369–74.

McDonald D, Price MN, Goodrich J *et al*. An improved Greengenes taxonomy with explicit ranks for ecological and evolutionary analyses of bacteria and archaea. *ISME J* 2012;**6**:610.

Moore TA, Xing Y, Lazenby B *et al*. Prevalence of anaerobic ammonium-oxidizing bacteria in contaminated groundwater. *Environ Sci Technol* 2011;**45**:7217–25.

Muyzer G, Dewaal EC, Uitterlinden AG. Profiling of complex microbial populations by denaturing gradient gel electrophoresis analysis of polymerase chain reaction-amplified genes coding for 16S rRNA. *Appl Environ Microbiol* 1993;**59**:695–700.

Narita Y, Zhang L, Kimura Z-i *et al*. Enrichment and physiological characterization of an anaerobic ammonium-oxidizing bacterium '*Candidatus* Brocadia sapporoensis'. *Syst Appl Microbiol* 2017;**40**:448–57.

Neef A, Amann R, Schlesner H *et al*. Monitoring a widespread bacterial group: *in situ* detection of planctomycetes with 16S rRNA-targeted probes. *Microbiology* 1998;**144**:3257–66.

Ni B-J, Hu B-L, Fang F *et al*. Microbial and physicochemical characteristics of compact anaerobic ammonium-oxidizing granules in an upflow anaerobic sludge blanket reactor. *Appl Environ Microbiol* 2010;**76**:2652–6.

Oshiki M, Shinyako-Hata K, Satoh H *et al*. Draft genome sequence of an anaerobic ammonium-oxidizing bacterium, "Candidatus *Brocadia sinica*". *Genome Announ* 2015;**3**:e00267–00215.

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Park H, Brotto AC, van Loosdrecht MCM *et al.* Discovery and metagenomic analysis of an anammox bacterial enrichment related to *Candidatus* "Brocadia caroliniensis" in a full-scale glycerol-fed nitritation-denitritation separate centrate treatment process. *Water Res* 2017;**111**:265–73.

Penton CR, Devol AH, Tiedje JM. Molecular evidence for the broad distribution of anaerobic ammonium-oxidizing bacteria in freshwater and marine sediments. *Appl Environ Microbiol* 2006;**72**:6829–32.

Qiao S, Bi Z, Zhou J *et al.* Long term effects of divalent ferrous ion on the activity of anammox biomass. *Bioresource Technol* 2013;**142**:490–7.

Schmid M, Twachtmann U, Klein M *et al.* Molecular evidence for genus level diversity of bacteria capable of catalyzing anaerobic ammonium oxidation. *Syst Appl Microbiol* 2000;**23**:93–106.

Schmid M, Walsh K, Webb R *et al.* *Candidatus* "Scalindua brodae", sp. nov., *Candidatus* "Scalindua wagneri", sp. nov., two new species of anaerobic ammonium oxidizing bacteria. *Syst Appl Microbiol* 2003;**26**:529–38.

Sonthiphand P, Neufeld JD. Evaluating primers for profiling anaerobic ammonia oxidizing bacteria within freshwater environments. *PLoS One* 2013;**8**:e57242.

Tsushima I, Kindaichi T, Okabe S. Quantification of anaerobic ammonium-oxidizing bacteria in enrichment cultures by real-time PCR. *Water Res* 2007;**41**:785–94.

van Niftrik L, Jetten MSM. Anaerobic ammonium-oxidizing bacteria: Unique microorganisms with exceptional properties. *Microbiol Mol Biol Rev* 2012;**76**:585–96.

Wang Q, Garrity GM, Tiedje JM *et al.* Naïve bayesian classifier for rapid assignment of rrna sequences into the new bacterial taxonomy. *Appl Environ Microbiol* 2007;**73**:5261–7.

Wang Z-b, Ni S-Q, Zhang J *et al.* Gene expression and biomarker discovery of anammox bacteria in different reactors. *Biochem Eng J* 2016;**115**:108–14.

Wenk CB, Blees J, Zopfi J *et al.* Anaerobic ammonium oxidation (anammox) bacteria and sulfide-dependent denitrifiers coexist in the water column of a meromictic south-alpine lake. *Limnol Oceanogr* 2013;**58**:1–12.

Yilmaz P, Parfrey LW, Yarza P *et al.* The SILVA and "All-species Living Tree Project (LTP)" taxonomic frameworks. *Nucleic Acids Res* 2014;**42**:D643–8.

Zhang L, Narita Y, Gao L *et al.* Maximum specific growth rate of anammox bacteria revisited. *Water Res* 2017;**116**:296–303.

Zheng B, Zhang L, Guo J *et al.* Suspended sludge and biofilm shaped different anammox communities in two pilot-scale one-stage anammox reactors. *Bioresource Technol* 2016;**211**:273–9:.

Zhou Z, Wei Q, Yang Y *et al.* Practical applications of PCR primers in detection of anammox bacteria effectively from different types of samples. *Appl Microbiol Biotechnol* 2018;**102**:5859–71.

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